WORKSHEET

Chapter 4: Multiplication in the plane

1 Find the resulting vector when each of the following is multiplied by the complex number i:

a $\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)$

b $\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)$

c $\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)$

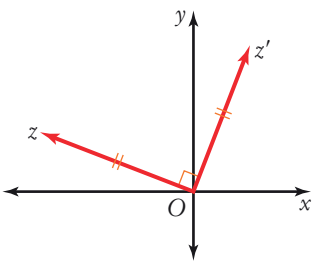
d $3\left[\cos\left(\frac{-\pi}{3}\right) + i\sin\left(\frac{-\pi}{3}\right)\right]$

e $\sqrt{6}\left[\cos\left(\frac{-3\pi}{4}\right) + i\sin\left(\frac{-3\pi}{4}\right)\right]$

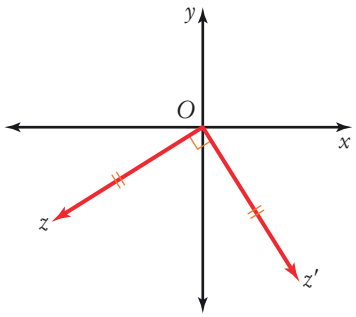
f $\sqrt{2}\left[\cos\left(\frac{-\pi}{6}\right) + i\sin\left(\frac{-\pi}{6}\right)\right]$

2 In each of the following, multiplication by which complex number takes z to z'?

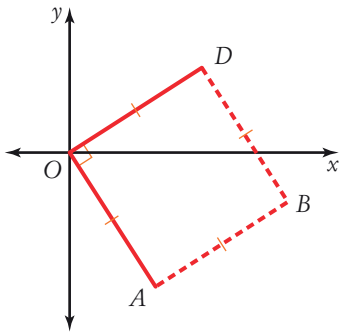
a



b



3 In the diagram, $OABC$ is a square. The point A represents the complex number a.

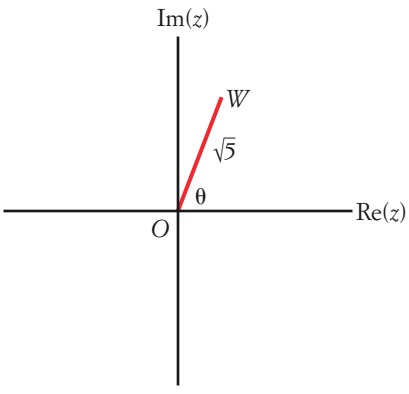


Find, in terms of a, the complex numbers represented by

a D

b B

4 The point W represents the complex number $w = \sqrt{5}\,\text{cis}(\theta)$ on an Argand diagram.



Sketch on the diagram provided:

a $-2w$

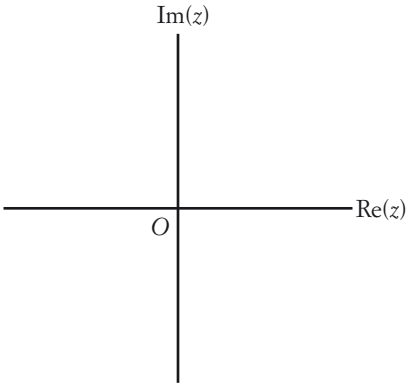
b $-iw$

c iw

d w^2

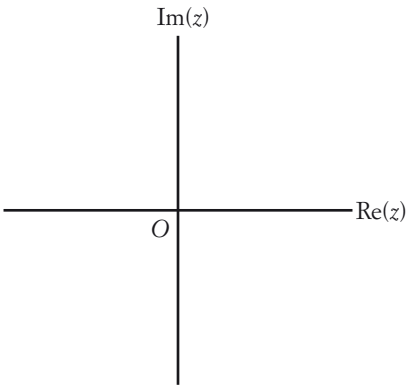
5 The complex numbers $z = 2\left[\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right]$ and $w = 3\left[\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right]$ are represented by the points Z and W on the complex plane. U represents the product zw.

Sketch the points Z, W and U on the complex plane provided.



6 The complex numbers $m = \sqrt{3}\left[\cos\left(\frac{-\pi}{3}\right) + i\sin\left(\frac{-\pi}{3}\right)\right]$ and $n = 3\left[\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right]$ are represented by the points M and N on the complex plane. P represents the product mn.

Sketch the points M, N and P on the complex plane provided.

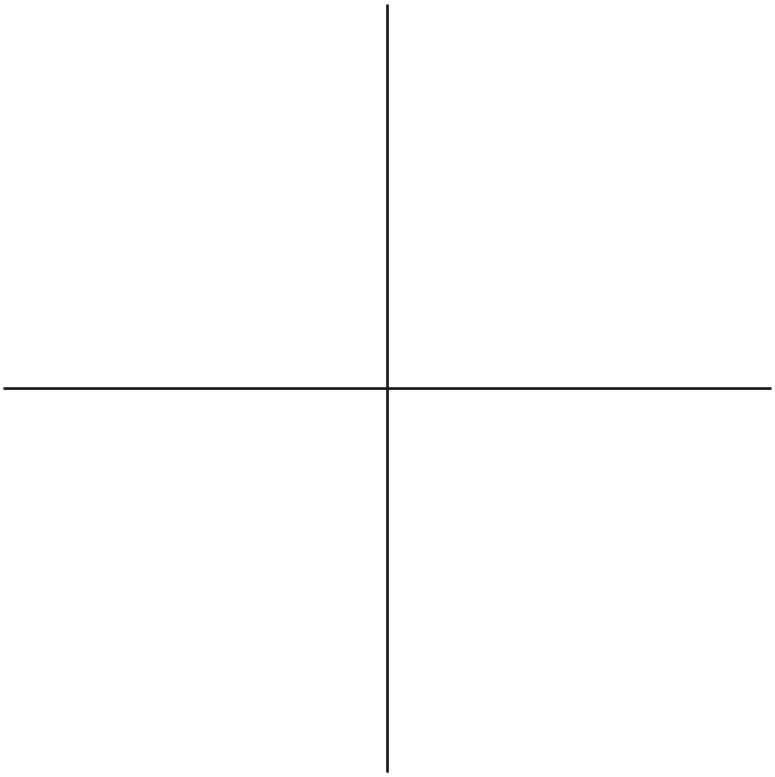


7 Consider the parallelogram $OPQR$, where P, Q and R represent the complex numbers p, q and r respectively. If $p = \cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)$, $OQ = 3OP$ and $\angle QOP = \frac{2\pi}{3}$, find the complex numbers q and p.

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8 Given that $p = \cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)$, sketch the following on an Argand diagram:

a p **b** p^2 **c** p^3
d p^4 **e** p^5 **f** p^6
g p^7 **h** p^8 **i** p^9
j p^{10} **k** p^{11} **l** p^{12}



Answers

1 **a** $\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)$ **b** $\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)$

c $\cos\left(\frac{-2\pi}{3}\right) + i\sin\left(\frac{-2\pi}{3}\right)$ **d** $3\left[\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right]$

e $\sqrt{6}\left[\cos\left(\frac{-\pi}{4}\right) + i\sin\left(\frac{-\pi}{4}\right)\right]$ **f** $\sqrt{2}\left[\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right]$

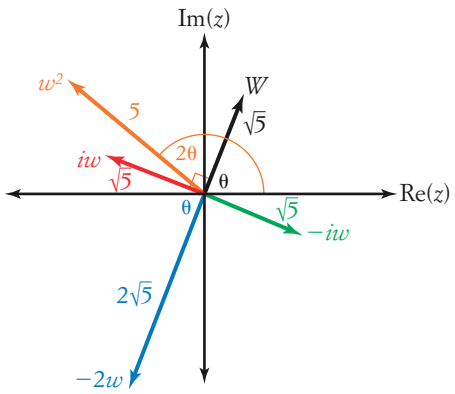
2 **a** $-i$

b i

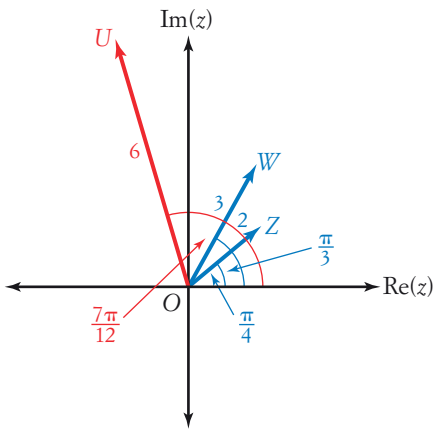
3 **a** ai

b $a + ai$

4

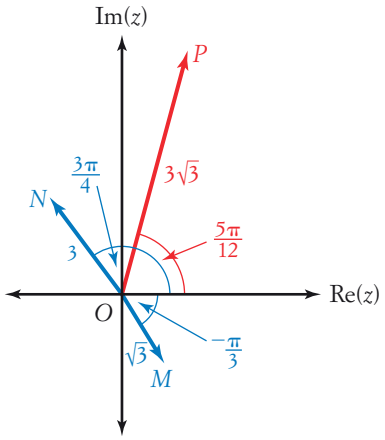


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6



7 $q = 3\left[\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)\right]$, $r = p + q = \text{cis}\left(\frac{\pi}{6}\right) + 3\ \text{cis}\left(\frac{5\pi}{6}\right)$

8

